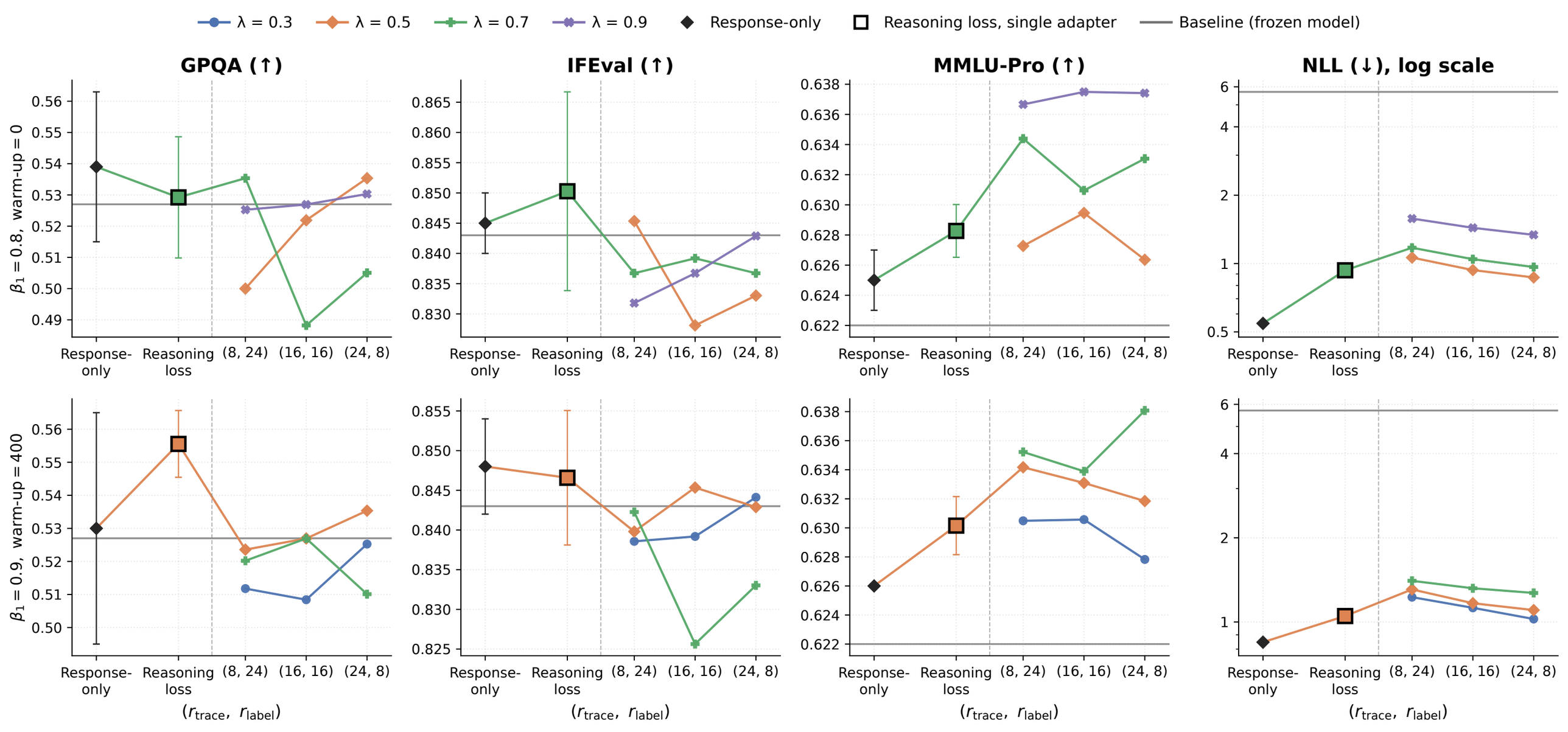
λ = 0.3
λ = 0.5
λ = 0.7
λ = 0.9
Response-only
Reasoning loss, single adapter
Baseline (frozen model)
GPQA (↑)
IFEval (↑)
MMLU-Pro (↑)
NLL (↓), log scale
$\beta_1 = 0.8$, warm-up = 0
$\beta_1 = 0.9$, warm-up = 400
Response-only
Reasoning loss
(8, 24)
(16, 16)
(24, 8)
$(r_{\text{trace}}, r_{\text{label}})$